

Master's Thesis

Circle Bundles with Positive Scalar Curvature over Enlargeable Manifolds

Johannes Nikolaus Vinnemeier

Supervisors: Prof. Dr. Bernhard Hanke,
Dr. Georg Frenck

Faculty of Mathematics, Natural Sciences and
Materials Engineering

University of Augsburg

Introduction

The scalar curvature of a Riemannian Manifold (M, g) is a smooth function $\text{scal}: M \rightarrow \mathbb{R}$ measuring how much the volume of small geodesic balls around a point deviates from that of euclidian balls. While it is the weakest curvature condition for a Riemannian manifold, the presence of positive scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet). *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \text{scal} \, dA = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

So any closed surface carrying a positive scalar curvature metric must be diffeomorphic to either the sphere S^2 or $\mathbb{R}P^2$. For non-compact complete manifolds, merely having positive scalar curvature may be too weak of a condition. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a positive scalar curvature metric with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One statement in this direction is the following theorem by Gromov:

Theorem (Enlargeable Decay Theorem, [?], page 15f.). *Let X be a complete Riemannian manifold, $E \rightarrow M$ a two dimensional vector bundle over an enlargeable manifold M . Assume that*

- (i) *There exists a proper map $f: X \rightarrow E$, $\deg(f) \neq 0$,*
- (ii) *The total space of the circle bundle corresponding to E is also enlargeable.*

Then

$$\min_{B_{x_0}(R)} \text{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$.

It is easy to see that condition ?? is always fulfilled if the bundle $E \rightarrow M$ is trivial (compare Lemma ?? ??), which lead Gromov to pose the following question:

Question ([?]). “What happens to nontrivial bundles $E \rightarrow M$? [...] In general, the examples in [?] indicate a possibility of non-enlargeable circle bundles over enlargeable M ; yet, it seems hard(er) to find such examples, where the corresponding total space would admit metrics with $\text{scal} > 0$.”

As it turns out, such examples do in fact exist, that is, circle bundles over enlargeable manifolds, where the total space admits a metric of positive scalar curvature and thus, in particular, is not enlargeable. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [?] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n} , $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M . In order to construct a positive scalar curvature metric on E , consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi: W \rightarrow \mathbb{R}$ with critical points of index at most n , which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Gromov-Lawson-Schoen-Yau Surgery Theorem (see Theorem ??).

In this thesis, we provide an alternative, much more direct construction relying simply on basic topological concepts. It should be said, however, that this alternative approach only provides examples in the totally non-spin case.

The thesis is organized as follows. Chapter 1 recalls some concepts from the broader theory of principal bundles, laying the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems for positive scalar curvature metrics, which are used in the main construction in Chapter 4. The proofs of these theorems are based on methods from surgery theory, which we also take some time to explain. Chapter 3 provides a brief discussion of the concept of enlargeability, which is the central idea underlying this thesis. Finally, Chapter 4 contains the main results. We present a simple construction of circle bundles over enlargeable manifolds whose total spaces admit metrics of positive scalar curvature in all dimensions ≥ 4 . These bundles are then used to obtain counterexamples to the Enlargeable Decay Theorem above if condition ?? is dropped.

1 Preliminaries

This chapter develops the necessary preliminaries for what follows, by recalling the notions of principal bundles, classifying spaces and characteristic classes. We assume the reader to be familiar with the concept of a fiber bundle with structure group. A classical reference for this is for example [?]. Moreover, we presuppose some basic familiarity with homotopy and homology theory.

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G -principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G -principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$ choose a local trivialisation

$$\phi: U \times G \rightarrow \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since by definition G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G -principal bundle $P \rightarrow B$ to an H -principal bundle $P' \rightarrow B'$ is a pair (F, f) of continuous maps $F: P \rightarrow P'$ and $f: B \rightarrow B'$ together with a homomorphism $\rho: G \rightarrow H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ -equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$ we assume ρ to be the identity and if $B = B'$ we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G -principal bundles over the same base space is an isomorphism.*

Proof. First suppose we are given a morphism

$$F: B \times G \rightarrow B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of equivariant maps $\phi_x: G \rightarrow G$. By equivariance $\phi_x(g) = \phi_x(e)g$ and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G -principal bundle is locally trivial this shows that we can always find local inverses in the general case and since inverses are unique they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G one can canonically associate a G -principal bundle, obtained by applying a general construction, which can be used to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p: E \rightarrow B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha: U_\alpha \times F \rightarrow p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$ be the corresponding transition functions defined by

$$(\phi_\beta^{-1} \circ \phi_\alpha)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) / \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha, \beta}(b) \cdot f', \alpha)$$

and let $p' : E' \rightarrow B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha : U_\alpha \times F' \rightarrow (p')^{-1}(U_\alpha), (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p' : E' \rightarrow B$ the structure of a fiber bundle with fiber F' and structure group G . We say that this new bundle is obtained from $E \rightarrow B$ by *changing the fiber* from F to F' (with the G -action on F' being implicit). Note that we can also obtain $E \rightarrow B$ from $E' \rightarrow B$ by changing the fiber from F' back to F . Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \rightarrow B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking if we want to understand the bundle $E \rightarrow B$ it is sufficient to understand its associated principal bundle, as it contains the same information as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example many constructions simplify for principal bundles, such as the concept of universal bundles as we will see in the next subsection. Another such instance is the construction for changing the fiber from above, which can be carried out more directly in the case of principal bundles and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p : P \rightarrow B$ be a G -principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F) / \sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$

together with the map $\pi : E \rightarrow B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G . Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial let $\psi : U \times G \rightarrow P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \text{id}_F : U \times G \times F \rightarrow p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa} : (U \times G \times F) / \sim \rightarrow (p^{-1}(U) \times F) / \sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover the map

$$U \times G \times F \rightarrow U \times F, (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F) / \sim \rightarrow U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$.

Under this homeomorphism $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa : U \times F \rightarrow \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$ and

$$\begin{array}{ccc} U \times F & \xrightarrow{\kappa} & \pi^{-1}(U) \\ & \searrow \text{proj.} & \downarrow \pi \\ & & U \end{array}$$

commutes. So $P \times_G F$ is locally trivial. To see that this construction agrees with the change of fiber construction from above if G acts effectively we have to compute the transition functions of the local trivialisations. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for P with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow G$. Moreover let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $P \times_G F$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([\psi_\beta(x, e), f]) = (x, \tau_{\alpha, \beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles with structure group are uniquely determined by the transition functions of local trivialisations. $\square$

In view of the above construction one usually denotes the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F as $P \times_G F \rightarrow B$. The Borel construction is natural in the following sense.

Lemma 1.5. *Let $p: P \rightarrow B$ be a G -principal bundle, F a space upon which G acts from the left and $f: B' \rightarrow B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\} / \sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\} / \sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic. $\square$

Lets look at some concrete examples for the theory developed above.

Example 1.6. For simplicity we assume all base spaces B to be smooth manifolds in this example.

- i) Let $E \rightarrow B$ be a real rank n vector bundle. Choosing a bundle metric gives $E \rightarrow B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $O(E)$ of E , whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x . Local trivialisations of the canonical map $p: O(E) \rightarrow B$ can be defined as follows. Choose an orthonormal local frame $e_1, \dots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi: U \times O(n) \rightarrow p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \dots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$, i.e. $\phi(x, A)$ is the basis of $E_x = p^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A . The transition functions of these trivialisations agree with the transition functions of the trivialisations of E . Again, since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principle bundle as defined at the beginning of this section.

- ii) An orientation of a real vector bundle allows us to further reduce the structure group to $SO(n)$. Similiarly, the associated principal bundle is then given by the *oriented orthonormal frame bundle* $SO(E)$ of E , whose fiber over a point $x \in B$ is the space of all oriented ordered orthonormal bases of E_x .
- iii) An example, which will be of relevance later is the following. Let $P \rightarrow B$ be an S^1 -principal bundle. We can change the fiber of P from S^1 to D^2 to obtain another bundle $E \rightarrow B$, where S^1 acts on D^2 by rotations. Since this action just extends the action of $S^1 = \partial D^2$ on itself we see that $P \cong \partial E$. So in particular the total space of any S^1 -principal bundle is null-bordant.

1.2 Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. Again, this classification problem turns out to be easier for principal bundles. For simplicity we assume all base spaces to be CW-complexes in this section.

Definition 1.7. A G -principal bundle $E \rightarrow B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G -principal bundles are G -homotopic if they are homotopic through morphisms of G -principal bundles.

Theorem 1.8. *Let $E \rightarrow B$ be a universal G -principal bundle. Then for any other G -principal bundle $P \rightarrow X$ there exists a morphism $P \rightarrow E$ of G -principal bundles, which is unique up to G -homotopy.*

Before showing the theorem let us discuss its implications. Firstly we can equivalently formulate the theorem by saying that $E \rightarrow B$ is a terminal object in the homotopy category of G -principal bundles over CW-complexes and thus in particular uniquely determined (up to G -homotopy equivalence). This justifies writing $EG := E$ and $BG := B$ and we refer to $EG \rightarrow BG$ as *the universal G -principal bundle*. The base space BG is called the *classifying space* of G . Since any bundle morphism $(F, f): P \rightarrow EG$ induces a morphism $P \rightarrow f^*EG$ (and vice versa, which is true for arbitrary fiber bundles) we conclude from Proposition ?? that

$$P \cong f^*EG.$$

We call f a *classifying map* for P . By homotopy invariance of the pullback we get a well-defined bijection

$$\phi_X: [X, BG] \rightarrow \mathcal{P}_G(X), [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G -principal bundles over X . To see that ϕ_X is indeed bijective note that the existence result in Theorem ?? shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover ϕ_X is clearly natural in X . Thus we have shown.

Corollary 1.9. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-): \text{HCW} \rightarrow \text{Set},$$

where HCW denotes the homotopy category of CW-complexes.

Theorem ?? will be a simple application of the following theorem.

Theorem 1.10. *Let $P \rightarrow X$ be a G -principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \rightarrow EG$ of G -principal bundles can be extended to a morphism $P \rightarrow EG$ of G -principal bundles.*

The proof of this Theorem based on an inductive cell by cell approach. We refer the reader to [?, Theorem 2.7.1].

Proof of Theorem ??. Applying Theorem ?? to $A = \emptyset$ shows the existence of a morphism $P \rightarrow EG$. Now suppose we are given morphisms $F_0, F_1: P \rightarrow EG$. Consider the G -principal bundle

$$\pi \times \text{id}: P \times [0, 1] \rightarrow X \times [0, 1],$$

where $\pi: P \rightarrow X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0, 1] \rightarrow X$). Then F_0 and F_1 define a morphism of G -principal bundles

$$(F_0, F_1): P \times \{0, 1\} = (P \times [0, 1])|_{X \times \{0, 1\}} \rightarrow EG,$$

which by Theorem ?? can be extended to $P \times [0, 1]$, thus yielding a G -homotopy between F_0 and F_1 . $\square$

Of course for this theory to be of any use we need such universal bundles to exist. The following general existence theorem is due to Milnor.

Theorem 1.11 (OG Milnor paper, Ebert). *For any topological group G there exists a universal G -principal bundle $EG \rightarrow BG$, where BG is a CW-complex.*

A priori the space BG constructed in [OG Milnor paper, Ebert] may not be a CW complex. However, if we choose a CW-approximation $f: X \rightarrow BG$ the total space of the bundle $f^*EG \rightarrow X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{array}{ccccccccc} \pi_{k+1}(X) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(f^*EG) & \longrightarrow & \pi_k(X) & \longrightarrow & \pi_{k-1}(G) \\ \downarrow \cong & & \downarrow \text{id} & & \downarrow & & \downarrow \cong & & \downarrow \text{id} \\ \pi_{k+1}(BG) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(EG) & \longrightarrow & \pi_k(BG) & \longrightarrow & \pi_{k-1}(G) \end{array}$$

We can now use the classification results for principal bundles to obtain analogous results for general fiber bundles. As discussed in section ?? we can generally recover information about arbitrary fiber bundles by changing the fiber.

Corollary 1.12. *Let G be a topological group acting effectively on a space F . Then for any CW-complex X the map*

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X .

Proof. By ?? the map

$$[X, BG] \rightarrow \mathcal{P}_G(X), \quad [f] \mapsto [f^*EG]$$

is a natural bijection and by Lemma ??

$$\mathcal{P}_G(X) \rightarrow \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^*EG) \times_G F].$$

So the statement follows by again applying Lemma ??. □

Again let us discuss a few examples.

Example 1.13.

- (i) The universal cover $\mathbb{R} \rightarrow S^1$ is a $\mathbb{Z}$ -principal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.
- (ii) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected since it is the image of the connected space EG under $EG \rightarrow BG$.

- (iii) The universal $O(k)$ -principal bundle $EO(k) \rightarrow BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n \times k}$, which is defined as the compact space of orthonormal k -tuples of vectors in $\mathbb{R}^n$. Moreover let Gr_k^n be the Grassmanian of k -planes in $\mathbb{R}^n$, i.e. the set of k -dimensional linear subspaces of $\mathbb{R}^n$. The topology on Gr_k^n is induced by the projection map

$$p: V_k^n \rightarrow \text{Gr}_k^n, \quad (v_1, \dots, v_k) \mapsto \text{span}(v_1, \dots, v_k)$$

Identifying a linear subspace $V \in \text{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V , providing local trivialisations for p . The transition of two such trivialisations is realized by actions of $O(k)$, hence p defines an $O(k)$ -principal bundle. Now the canonical inclusion $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$ induces inclusions $V_k^n \rightarrow V_k^{n+1}$ and $\text{Gr}_k^n \rightarrow \text{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \text{colim}_n (V_k^n), \quad \text{Gr}_k^\infty := \text{colim}_n (\text{Gr}_k^n)$$

and the induced map $V_k^\infty \rightarrow \text{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \rightarrow \text{Gr}_k^n$ for varying n , which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \rightarrow \text{Gr}_k^\infty$ defines an $O(k)$ -principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \rightarrow S^{n-1}, (v_1, \dots, v_k) \mapsto v_1.$$

Similar arguments as above shows that this map is locally trivial with fiber V_{k-1}^{n-1} . Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_k^n) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1} we conclude that the image of any map $S^i \rightarrow V_k^\infty$ is contained in some finite stage V_k^n . Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$ -principal bundle is explicitly given by $V_k^\infty \rightarrow \text{Gr}_k^\infty$.

Similarly we get an explicit description of the universal $SO(k)$ -principal bundle. Let $\tilde{\text{Gr}}_k^n$ be the set of tuples consisting of a k -dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\text{Gr}}_k^n$ is induced by

$$p: V_k^n \rightarrow \tilde{\text{Gr}}_k^n, (v_1, \dots, v_k) \mapsto (\text{span}(v_1, \dots, v_k), \text{o}_{(v_1, \dots, v_k)}),$$

where $\text{o}_{(v_1, \dots, v_k)}$ is the unique orientation on $\text{span}(v_1, \dots, v_k)$, such that $(v_1, \dots, v_k)$ is positively oriented. Similar to above we see that p defines an $SO(k)$ -principal bundle. Taking the colimit, we obtain the universal $SO(k)$ -principal bundle $V_k^\infty \rightarrow \tilde{\text{Gr}}_k^\infty$.

- (iv) Any real k -dimensional vector bundle $E \rightarrow X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary ?? and the example above show that every such E is the pullback of the universal real k -dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \rightarrow \text{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \rightarrow \text{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X, v) \in \text{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞ . An element in V_k^∞ is a k -tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k, (A, v) \mapsto (\text{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \rightarrow \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similarly, oriented real k -dimensional vector bundles are classified by maps to $\tilde{\text{Gr}}_k^\infty$. We claim that the universal oriented real k -dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \rightarrow \tilde{\text{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, \text{o}_X, v) \in \tilde{\text{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\text{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k^{SO}, (A, v) \mapsto (\text{im}(A), \text{o}_A, Av),$$

where o_A is the unique orientation on $\text{im}(A)$, such that $A: \mathbb{R}^k \rightarrow \text{im}(A)$ is orientation preserving. Again this map descends to the quotient defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \rightarrow \gamma_k^{SO}$.

- (v) Completely analogous to the real case we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmanians $\text{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$ -principal bundle is given by $V_k^\infty(\mathbb{C}) \rightarrow \text{Gr}_k^\infty(\mathbb{C})$. The universal complex k -dimensional vector bundle is the tautological bundle $\tau_k \rightarrow \text{Gr}_k^\infty(\mathbb{C})$.

- (vi) As special cases of the preceding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3 Stiefel-Whitney and Chern-Classes

In the preceeding Chapter we discussed the question of how bundles can be classified. Another fundamental question is of course how to determine, whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by defining computable invariants. For vector bundles the most important invariants are so called *characteristic classes*. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\text{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$ -vector bundles over a space B . Then $\text{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\text{Top} \rightarrow \text{Set}$. A characteristic class c for $\mathbb{K}$ -vector bundles with values in the cohomology theory H^* is a natural transformation from the functor $\text{Vect}_{\mathbb{K}}(-)$ to the functor H^* regarded as a functor with values in Set . So in more concrete terms c assigns to each $\mathbb{K}$ -vector bundle $E \rightarrow B$ a cohomology class $c(E) \in H^*(B)$ depending only on the isomorphism class of E , such that for any continuous map $f: B' \rightarrow B$ the following naturality condition holds

$$c(f^*E) = f^*c(E).$$

Therefore, using the theory of classifying spaces developed above, we see that for a real (resp. complex) k -dimensional vector bundles, the characteristic class c is uniquely determined by its value for the universal bundle γ_k (resp. τ_k) discussed in example ?? ?? (resp. ??).

For real vector bundles the most important characteristic classes are the *Stiefel-Whitney classes* and for complex vector bundles the *Chern classes*. Both are formally quite similar and can be characterized by a few axioms.

Definition 1.14 (Stiefel-Whitney Classes, HATCHER). There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

- i) $w_i(E) \in H^i(B; \mathbb{Z}_2)$ for any real bundle $E \rightarrow B$,
- ii) $w_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,
- iv) For the canonical line bundle $\gamma_1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property ?? the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class* of E . We sometimes write $w_0(E) := 1 \in H^0(B; \mathbb{Z}_2)$ to simplify notation.

Definition 1.15 (Chern Classes, HATCHER). There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

- i) $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for any complex bundle $E \rightarrow B$,
- ii) $c_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,
- iv) For the canonical line bundle $\tau_1 \rightarrow \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(B; \mathbb{Z})$. We refer the reader to [HATCHER] for a proof of existence and uniqueness of the Stiefel-Whitney and Chern classes.

Example 1.16.

- i) Let $\iota: \mathbb{R}P^n \rightarrow \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \rightarrow \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology we conclude from naturality that $w_1(\gamma_1^n) = \iota^*w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly, $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.
- ii) Let $\varepsilon \rightarrow B$ be a trivial real bundle over some space B . Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f: B \rightarrow B$ and thus $w_i(\varepsilon_n) = f^*w_i(\varepsilon) = 0$ for all $i \geq 1$. Similarly all Chern classes for complex trivial bundles vanish.

- iii) Let $TS^n \rightarrow S^n$ be the tangent bundle of the sphere and let $\nu_n \rightarrow S^n$ be the normal bundle with respect to the standard embedding $S^n \rightarrow \mathbb{R}^{n+1}$. Since ν_n and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property ?? in Definition ?? that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we saw that although all Stiefel-Whitney classes of a bundle vanish, it need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that the first Stiefel-Whitney (resp. Chern) class already completely determines the bundle:

Theorem 1.17 (Hatcher, Proposition 3.10). *Let X be a CW-complex. Then the maps*

$$w_1: \text{Vect}_{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}_2)$$

and

$$c_1: \text{Vect}_{\mathbb{C}}(X) \rightarrow H^2(X; \mathbb{Z})$$

are bijections.

While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orientability:

Proposition 1.18 (Hatcher, Prop 3.11). *Let X be a CW-complex. A real vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.*

Similarly vanishing of the second Stiefel Whitney class corresponds to the existence of another structure, namely that of a *spin structure*.

Definition 1.19. (i) For $n \geq 2$ the *spin group* $\text{Spin}(n)$ is defined as the total space of the non-trivial two-sheeted cover $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$. If $n \geq 3$ this is the universal cover and in the case $n = 2$ we have $\text{Spin}(2) \cong \text{SO}(2)$. For $n = 1$ we define $\text{Spin}(1) := \mathbb{Z}_2$, the total space of the two-sheeted cover $\rho: \mathbb{Z}_2 \rightarrow \text{SO}(1)$.

- (ii) Let $E \rightarrow B$ be an oriented real vector bundle equipped with a fiber metric and let $P_{\text{SO}(n)}$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\text{Spin}(n)$ -principal bundle $P_{\text{Spin}(n)}$ together with a map

$$\phi: P_{\text{Spin}(n)} \rightarrow P_{\text{SO}(n)}$$

of principal bundles, which is ρ -equivariant (compare Definition ??).

- (iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Remark 1.20. When $n = 1$, $P_{\text{SO}(1)} \cong X$ and a spin structure is simply a two-sheeted cover of X .

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric on E , since different choices yield isomorphic oriented orthonormal frame bundles. In fact the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above.

Theorem 1.21 (Lawson, Micheaslon). *Let X be a CW-complex. An orientable real vector bundle $E \rightarrow X$ admits a spin structure if and only if $w_2(E) = 0$.*

So for example all spheres S^n are spin (see example ??). A non-example for a spin manifold is $\mathbb{C}P^2$ (see [MilnorStasheff??]).

Remark 1.22. Let $E \rightarrow X$ be a real vector bundle over a CW-complex. If an orientation is given, E is classified by a map $X \rightarrow B\text{SO}(n)$. Similarly, if a spin structure exists, E is classified by a map $X \rightarrow B\text{Spin}(n)$, which can be seen as follows. Let $\phi: P_{\text{Spin}(n)} \rightarrow P_{\text{SO}(n)}$ be a spin structure on E and let $f: X \rightarrow B\text{Spin}(n)$ be a classifying map for $P_{\text{Spin}(n)}$. Define a vector bundle

$$\gamma_n^{\text{Spin}} := E\text{Spin}(n) \times_{\text{Spin}(n)} \mathbb{R}^n$$

using Lemma ??, where $\text{Spin}(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$ and the canonical action of $\text{SO}(n)$. From Lemma ?? we obtain

$$f^* \gamma_n^{\text{Spin}} \cong P_{\text{Spin}(n)} \times_{\text{Spin}(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $P_{\text{Spin}(n)} \times_{\text{Spin}(n)} \mathbb{R}^n \rightarrow P_{\text{SO}(n)} \times_{\text{SO}(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

This map is clearly a linear bijection on fibers, hence an isomorphism of vector bundles.

Recall that a manifold M (only if $\dim \geq 3$ or so??) is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words “ TM does not contain any Moebius Bands”. A similiar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.23 (Georg Paper, Proposition 2.6). *Let M be a connected oriented manifold of dimension at least 5.*

- i) *M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.*
- ii) *M is totally non-spin if and only if there exists an embedded sphere $S^2 \subseteq M$, such that the restriction $TM|_{S^2}$ of the tangent bundle to this sphere is non-trivial.*

Proof. Since we will not be making use of the first characterization, we shall only proof part ?? of the proposition and refer the reader to [Georg Paper] for a proof of part ??. Let $\tilde{M}$ be the universal cover of M . Since $T\tilde{M}$ does not admit a spin structure we conclude from Theorem ?? that $w_2(T\tilde{M}) \neq 0$. So by the universal coefficient theorem there exists a homology class $a \in H_2(\tilde{M}; \mathbb{Z}_2)$ with

$$\langle w_2(T\tilde{M}), a \rangle \neq 0.$$

Once more applying the universal coefficient theorem as well as the Hurewicz theorem we obtain an isomorphism

$$\pi_2(\tilde{M}) \otimes \mathbb{Z}_2 \rightarrow H_2(\tilde{M}; \mathbb{Z}) \otimes \mathbb{Z}_2 \cong H_2(\tilde{M}; \mathbb{Z}_2).$$

Explicitly, for some class $\alpha \in \pi_2(\tilde{M})$ represented by a map $f: S^2 \rightarrow \tilde{M}$ the above isomorphism maps $\alpha \otimes 1$ to $f_*[S^2]_{\mathbb{Z}_2}$, where $[S^2]_{\mathbb{Z}_2}$ is the $\mathbb{Z}_2$ -fundamental class of S^2 . Therefore we can write $a = f_*[S^2]_{\mathbb{Z}_2}$ for some $f: S^2 \rightarrow \tilde{M}$. Let $p: \tilde{M} \rightarrow M$ be the projection and define $g := p \circ f: S^2 \rightarrow M$. We compute

$$\langle w_2(g^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle f^*w_2(p^*TM), [S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), f_*[S^2]_{\mathbb{Z}_2} \rangle = \langle w_2(T\tilde{M}), a \rangle \neq 0,$$

where we used $p^*TM \cong T\tilde{M}$ and naturality of the Kronecker pairing in the second equation. As $\dim(M) \geq 5$ we may assume g to be an embedding and since $w_2(g^*TM) \neq 0$ by the above computation we conclude that g^*TM can not be trivial. $\square$

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

Lemma 1.24. *Let $E \rightarrow S$ be a real vector bundle of rank at least 3 over a two dimensional CW-complex S . Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem ?? and Remark ?? we conclude that there exists a classifying map $f: S \rightarrow B\text{Spin}(n)$ for E , where n is the rank of E . From Example ?? ?? it follows that $\pi_k(B\text{Spin}(n)) = 0$ for $k \geq 2$. Since S is two dimensional the map f is homotopic to a constant ([Hatcher, Lemma 4.6]), therefore $E \cong f^*\xi_{\text{Spin}}$ is trivial. $\square$

2 Positive Scalar Curvature

The purpose of this chapter is twofold. The first half provides a brief introduction to scalar curvature, following [?] and [?]. In the second half, we prove two concrete existence results for metrics of positive scalar curvature, which will be used in the construction in Chapter ??.

2.1 A brief introduction

Let M^n be an n -dimensional manifold. Given any Riemannian metric g on M , we can associate to it a smooth function $\text{scal}_g: M \rightarrow \mathbb{R}$ called the *scalar curvature* of M . In the case where M is an embedded surface in $\mathbb{R}^3$, the scalar curvature equals twice the classical Gaussian curvature $K: M \rightarrow \mathbb{R}$. So in this case we can intuitively quantify the scalar curvature at a given point.

../graphics/scalar_curvature.pdf

In dimension 2 the scalar curvature is the only intrinsic notion of curvature, however in dimensions $n \geq 3$ there exists a whole hierarchy of different curvatures. The most prominent are sectional curvature, Ricci curvature and scalar curvature, where the sectional curvature is the strongest notion, determining the entire geometry of M . In this thesis we are concerned with the scalar curvature, which is the weakest of the three. In most textbooks on Riemannian geometry it is defined as twice the trace of the Riemannian curvature tensor. While this definition is best suited for explicit computations it provides no further intuition on what scalar curvature is. Since we have no need for doing such explicit computations, we give a more geometric definition, which essentially extends our intuition in the 2-dimensional case to arbitrary dimensions.

Definition 2.1. Let (M^n, g) be an n -dimensional Riemannian manifold. The scalar curvature $\text{scal}_g(x) \in \mathbb{R}$ at the point $x \in M$ is defined as the value appearing in the third coefficient of the Taylor expansion

$$\frac{\text{vol}(B_r(x))}{\text{vol}(B_r^{\text{eucl.}}(0))} = 1 - \frac{\text{scal}_g(x)}{6(n+2)}r^2 + \dots,$$

where $\text{vol}(B_r(x))$ denotes the volume of a geodesic ball in M of radius r centered at x and $\text{vol}(B_r^{\text{eucl.}}(0))$ denotes the volume of a ball of the same radius r in standard euclidian space $\mathbb{R}^n$.

Broadly speaking, the study of scalar curvature falls into two categories: *Rigidity* results and *flexibility* results. As the title of this chapter suggests, one is primarily concerned with results about *positive* scalar curvature. The reason for this is the following theorem.

Theorem 2.2. Let M^n be a closed manifold of dimension $n \geq 3$ and let $f: M \rightarrow \mathbb{R}$ be a smooth function.

- (i) If there exists a point $p \in M$ with $f(p) < 0$, then M admits a metric g with scalar curvature $\text{scal}_g = f$.
- (ii) If M admits a metric whose scalar curvature is non-negative but not identically zero then M admits another metric g with scalar curvature $\text{scal}_g = f$.

So part ?? of the theorem asserts that the only scalar curvature condition from which we can expect any kind of obstruction results is non-negativity. Moreover, aside from the scalar-flat case (where the scalar curvature is identically zero), part ?? shows that studying non-negative scalar curvature yields nothing new beyond what is already captured by positive scalar curvature. In the following we refer to a metric g on M , whose scalar curvature is everywhere positive as a PSC-metric.

The reason why the study of positive scalar curvature has attracted particular attention compared to other other curvature conditions, such as positive Ricci or sectional curvature, is that it strikes an exceptionally good balance between the rigidity and flexibility aspects. While positive Ricci or sectional curvature have substantial topological implications, this also makes it

significantly more difficult to construct metrics with these curvature conditions. For example the Bonnet-Myers theorem states that a positive lower bound on the Ricci curvature of M already implies that M must be compact with finite fundamental group. For PSC-metrics however, there are existence results under quite general assumptions, which we will discuss below. First let us consider some examples illustrating the rigidity aspects of positive scalar curvature. The most prominent obstruction to the existence of a PSC-metric comes from spin geometry. Let (M^n, g) be a closed Riemannian spin manifold with n even. Then the spin structure and the metric g determine a Hermitian vector bundle $S \rightarrow M$ together with a compatible connection ∇^S and a self-adjoint first order differential operator $D: \Gamma(S) \rightarrow \Gamma(S)$ acting on sections of S . Moreover, there exists a canonical splitting $S \cong S^+ \oplus S^-$ with respect to which D splits as

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}.$$

We call $S \rightarrow M$ the *spinor bundle* of M and D the *Dirac operator*. One of the central motivations behind constructing this operator was to find a first order differential operator squaring to the Laplacian. The spinor bundle S carries additional structure making it possible to define such an operator, however, its square agrees with the Laplacian only up to an error term determined by the curvature of M . More precisely, the *Schrödinger–Lichnerowicz formula* holds:

$$D^2 = \Delta^S + \frac{1}{4} \text{scal}_g,$$

where $\Delta^S: \Gamma(S) \rightarrow \Gamma(S)$ is the connection Laplacian determined by ∇^S . Given any vector bundle $E \rightarrow N$ over a closed Riemannian manifold N carrying a fiber metric and a compatible connection, it is a general fact that the induced connection Laplacian is a non-negative operator with respect to the L^2 -inner product $\langle \cdot, \cdot \rangle$ on $\Gamma(E)$ induced by the fiber metric and the volume form of N . Applying this to S , let $\psi \in \Gamma(S)$. Then

$$0 \leq \langle D\psi, D\psi \rangle = \langle D^2\psi, \psi \rangle = \langle \Delta^S\psi, \psi \rangle + \frac{1}{4} \int_M \text{scal}_g \|\psi\|^2,$$

where we used self-adjointness of D in the first equation and the Schrödinger–Lichnerowicz formula in the second. If $\text{scal}_g > 0$ and $\psi \neq 0$, the right-hand side is strictly positive, since Δ^S is a non-negative operator. But the right-hand side equals $\langle D\psi, D\psi \rangle = \|D\psi\|^2$, so this forces $D\psi \neq 0$. Hence the kernel of D must be trivial. In particular the integer

$$\text{ind}(D) := \dim(\ker D^+) - \dim(\ker D^-),$$

known as the *index* of D , must vanish (one can show that the kernels of $D^\pm$ are finite-dimensional, so this is well defined). Now the major insight is that this index can also be computed in terms of a purely topological invariant of M , namely the so-called $\hat{A}$ -class $\hat{A}(TM) \in H^{4*}(M; \mathbb{Q})$ of the tangent bundle $TM \rightarrow M$, which is a certain characteristic class. The connection between the index of the Dirac operator and the $\hat{A}$ -class of the tangent bundle is established by the famous *Atiyah–Singer index theorem*, which states:

$$\text{ind}(D) = \langle \hat{A}(TM), [M] \rangle,$$

where the right-hand side is the evaluation of the cohomology class $\hat{A}(TM)$ against the fundamental class $[M]$ of M (for the components of $\hat{A}(TM)$ in degree $\neq n$ this is 0, so in particular $\hat{A}(M) = 0$ if n is not divisible by 4). The number $\hat{A}(M) := \langle \hat{A}(TM), [M] \rangle \in \mathbb{Q}$ is also referred to as the $\hat{A}$ -genus of M . In summary:

Theorem 2.3 (Lichnerowicz, 1963). *Let (M^{4k}, g) be a closed Riemannian spin manifold with positive scalar curvature. Then $\hat{A}(M) = 0$.*

A further, more accessible and geometric example for a rigidity result concerning positive scalar curvature is the following classical result by Llarull.

Theorem 2.4 (Llarull, 1998). *Let M^n be a closed connected Riemannian spin manifold of dimension $n \geq 2$. Suppose $\text{scal}_g \geq n(n-1)$ and let $f: M \rightarrow S^n$ be a smooth 1-Lipschitz map with $\deg(f) \neq 0$. Then f is an isometry.*

The constant $n(n-1)$ appearing is no coincidence as the scalar curvature of the n -sphere S^n with the standard metric g_0 can be computed to precisely equal this constant. Llarull applied this theorem in the special case $M = S^n$ and $f = \text{id}$ to show that the only metric g on S^n with $\text{scal}_g \geq n(n-1)$ and $g \geq g_0$ is the standard metric. Said differently, one cannot increase the curvature of the sphere without contracting distances. We conclude our discussion of scalar curvature rigidity results with another quite geometric statement, which is an example for a so called *band width estimates*.

Theorem 2.5 (Gromov). *Let $M = T^{n-1} \times [-1, 1]$ and let g be a Riemannian metric on M with scalar curvature bounded from below by a positive constant $\sigma > 0$. Then the distance between the boundary components $\partial_- M = T^{n-1} \times \{-1\}$ and $\partial_+ M = T^{n-1} \times \{1\}$ satisfies*

$$\text{dist}_g(\partial_- M, \partial_+ M) \leq 2\pi \sqrt{\frac{n-1}{n\sigma}}.$$

Intuitively we think of M as a cylinder (a ‘band’) over T^{n-1} with a left boundary component $\partial_- M$ and a right boundary component $\partial_+ M$. Thus this theorem says, that the boundaries of a band, which is highly positively curved must be close together. There are various generalisations of Theorem ?? to allow for bands over other manifolds [?]. Gromov even conjectures that T^{n-1} can be replaced by any manifold not admitting a PSC-metric [Conjecture C in Metric inequalities].

Let us now turn to the flexibility aspect of positive scalar curvature. The primary method for constructing PSC-metrics is the following theorem, known as the *Surgery Theorem*, which has been shown independently by Schoen and Yau [SY79] as well as Gromov and Lawson [GL80]. We refer the reader to the Appendix for a brief introduction to surgery and cobordism theory.

Theorem 2.6 (Gromov, Lawson, 1980, Schoen Yau 1979, Chernysh Theorem Georg Paper). *Let M be a compact manifold, which admits a PSC-metric. Then any manifold M' obtained from M by surgery of codimension at least 3 also admits a PSC-metric.*

We remark that the construction of the PSC-metric on M' only changes the given PSC-metric on M locally near the embedding along which the surgery is performed. In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has for example been studied quite thoroughly in the context of the h -cobordism theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 2.7 (Wall, 1971). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion is r -connected for $r \leq \dim(W) - 4$. Then W contains only handles of index $< \dim(W) - r$.*

Combining the above theorem with the Surgery Theorem ??, we can make the following observation. If M_0 admits a psc metric and the inclusion $M_1 \rightarrow W$ is 2-connected, then M_1 also admits a psc metric. In fact, using the conclusion of Theorem ?? it is possible to obtain an even stronger result. We say that a metric g on a manifold M with boundary is of *product form near the boundary*, if there is some collar neighbourhood $\iota: \partial M \times [0, 1) \rightarrow M$, such that $\iota^*g = g|_{\partial M} + dt^2$, where $g|_{\partial M}$ is the induced metric on ∂M .

Theorem 2.8 (extending psc to cobordism, Georg, Gaja). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W$ is 2-connected. Then any psc metric on M_0 can be extended to a psc metric on W , which is of product form near the boundary.*

In view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is.

2.2 Existence results for PSC-metrics

We will prove the following two existence results for PSC-metrics, one in the spin case and one in the totally non-spin case.

Theorem 2.9. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

Theorem 2.10. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

We first show the totally non-spin case.

Proof of Theorem ??. Without loss of generality assume W to be connected as well (otherwise only consider the connected component containing M). We first show that $\pi_2(W, M)$ is finitely generated. Let $\tilde{M}$ and $\tilde{W}$ be the universal covers of M and W respectively. Then the inclusion $M \rightarrow W$ lifts to a map $\tilde{M} \rightarrow \tilde{W}$ by virtue of which we can consider the relative homotopy group $\pi_2(\tilde{W}, \tilde{M})$. It follows from the Hurewicz Theorem that

$$\pi_2(\tilde{W}, \tilde{M}) \cong H_2(\tilde{W}, \tilde{M}).$$

By ??? $H_2(\tilde{W}, \tilde{M})$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ module (??) hence the same holds for $\pi_2(\tilde{W}, \tilde{M})$. Now consider the following diagram where the rows are exact.

$$\begin{array}{ccccc} \pi_2(\tilde{M}) & \longrightarrow & \pi_2(\tilde{W}) & \twoheadrightarrow & \pi_2(\tilde{W}, \tilde{M}) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \\ \pi_2(M) & \longrightarrow & \pi_2(W) & \twoheadrightarrow & \pi_2(W, M) \end{array}$$

The two rightmost horizontal maps are surjective by exactness, since $\pi_1(\tilde{M}) = 0$ and $\pi_1(M) \rightarrow \pi_1(W)$ is an isomorphism. A diagram chase now shows that $\pi_2(\tilde{W}, \tilde{M}) \rightarrow \pi_2(W, M)$ is an isomorphism.

Next find lifts with... Let

$$F: W \rightarrow B := BSO(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i$, where $i: M_1 \rightarrow W$ is the inclusion. Then f is a classifying map for $TM_1 \oplus \mathbb{R}$ with $\mathbb{R}$ the trivial line bundle. We claim that f is surjective on π_2 . Since $\pi_2(B) \cong \pi_1(SO(n+1)) \cong \mathbb{Z}_2$ it suffices to show that $\pi_2(f)$ is non-zero. By Proposition ?? there exists an embedding $\iota: S^2 \rightarrow M_1$, such that $w_2(\iota^*TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^* \gamma_n^{\text{SO}}) = w_2(\iota^*(TM_1 \oplus \mathbb{R})) = w_2(\iota^*TM_1) \neq 0.$$

Therefore $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$ showing the claim. Now consider the following diagram, where the row is exact.

$$\begin{array}{ccccc} \pi_2(M_1) & \xrightarrow{i_*} & \pi_2(W) & \longrightarrow & \pi_2(W, M_1) \\ & \searrow f_* & \downarrow F_* & & \\ & & \pi_2(B) & & \end{array}$$

Using surjectivity of f_* , a diagram chase shows that we can find lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_* . Moreover, since $n \geq 5$ the classes α_i can be represented by embeddings

$$\phi_i: S^2 \rightarrow W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented it follows from Proposition ?? that $w_1(\phi_i^*TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, such that

$$w_2(\phi_i^*TW) = w_2((F \circ \phi_i)^* \gamma_n^{\text{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example ?? ??) we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^*TW) = 0$$

for $k = 1, 2$. Thus Lemma ?? shows that ν_i is indeed trivial. This allows us to perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$, see Proposition ?? ???. In other words, the inclusion $M_1 \rightarrow W'$ is 2-connected. So by Theorem ?? there exists a psc metric g on W' , which is of product form near the boundary. To construct a psc metric on the original cobordism W , observe that since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore the surgery theorem ?? applies to construct a psc metric on W . Since this construction only changes the given metric g locally near the surgery embeddings, the resulting metric on W is still of product form near the boundary. $\square$

Proof of Theorem ??. Again $\pi_2(W, M)$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ -module. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ be such generators. We could argue as in the proof of Theorem ??, replacing $BSO(n+1)$ by $B\text{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 as $\pi_2(B\text{Spin}(n+1)) \cong \pi_1(\text{Spin}(n+1)) = 0$. However this just amounts to the following more direct argument. Choose lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ and represent α_i by an embedding $\phi_i: S^2 \rightarrow W$. Then

$$w_k(\phi_i^*TW) = \phi_i^*w_k(TW) = 0$$

for $k = 1, 2$ since W is spin. The rest of the proof is now as in Theorem ??. $\square$

Remark 2.11. The similarity between the proofs of Theorems ?? and ?? is no coincidence, since both statements are a consequence of a more general result. This unifying approach can be formulated using the language of tangential 2-types, which we shall not discuss further in this thesis. See for example [Georg Paper etc.].

In the main construction in Chapter 4 we will invoke Theorem ?? to construct a PSC-metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 2.12. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

(ii) *A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.*

(iii) *If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .*

Remark 2.13. All statements of the above lemma remain valid if “totally non-spin” is replaced by “non-spin”. The corresponding proofs are just simpler versions of the arguments given below.

Proof. For ?? let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^*w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^*w_2(\pi_1^*T\tilde{M}_1 \oplus \pi_2^*T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^*T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^*T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^*T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For ?? suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M . Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{array}{ccc} T\tilde{N} & \xrightarrow{d\tilde{\iota}} & T\tilde{M} \\ \downarrow & & \downarrow \\ \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \end{array}$$

Therefore $\tilde{\iota}^*w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For ?? assume without loss generality that M is connected. Denote by $B^n := \text{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in ?? by noting that for a manifold X with boundary the universal cover of $\text{int}(X)$ is given by $\text{int}(\tilde{X})$ and the inclusion $\text{int}(\tilde{X}) \rightarrow \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p: \tilde{M} \rightarrow M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p . From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber F is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi: S^1 \rightarrow \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi: D^2 \rightarrow \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic rel S^1 to some map $\Phi_1: D^2 \rightarrow \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r: \tilde{M} - p^{-1}(0) \rightarrow \hat{M}_0$$

be such a retraction and let

$$r_t: \tilde{M} - p^{-1}(0) \rightarrow \tilde{M} - p^{-1}(0)$$

be a homotopy rel S^1 from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy rel S^1 from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{array}{ccc} S^{n-1} \times F & \longrightarrow & D^n \times F \\ \downarrow & & \downarrow \\ \tilde{M} - (B^n \times F) & \xrightarrow{\iota} & \tilde{M} \end{array}$$

where all maps are inclusions, then shows that ι is injective on $H^2(-; \mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^*T\tilde{M}) = \iota^*w_2(T\tilde{M}) \neq 0.$$

□